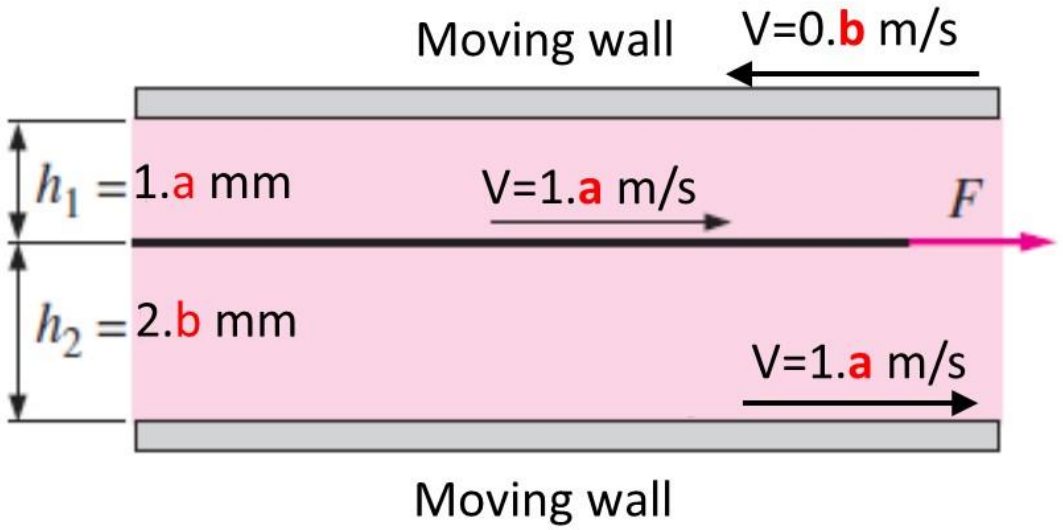
2020-2021 SPRING SEMESTER FLUID MECHANICS FINAL EXAM, Date:15/06/2021

Name: **Surname:** **Student ID:**

Question 1 (20p):

A flat plate which has dimensions of (3**a** cm x 3**a** cm) is pulled with a velocity of 1.**a** m/s horizontally inside an oily layer between two parallel walls, one moving with a constant velocity of 1.a m/s which has the same direction with the plate and the other moving with a constant velocity of 0.**b** m/s which has opposite direction as shown in the figure. The dynamic viscosity of the oil is 0.002**b** Pa.s. Assuming the velocity between two walls changes **linearly**, determine the force (F) needed to be applied on the plate to keep this motion stable.
Draw the velocity and shear stress distributions on both sides.



a, and b are the last two digits of your student ID.
For example, Student ID: 17042082 a=8 b=2